

Why open source?

One of the primary reasons for not securing networks is the expense involved. Here, open source software is an alternate to commercially available products. These include operating systems, word processors, spreadsheets, presentation tools, graphic tools, browsers, etc.

Open source has a wide range of firewall distributions to offer. Some of them are pfSense,Endian, Untangle, IPfire and, of course, IPCop. These can be installed on any computer system supported by the corresponding distro – the hardware configuration requirement is absolutely minimal. Even an old computer system with a Celeron CPU, 512 MB RAM and a 20 GB hard disk can run these firewalls. For the few who think open source has no support, some of these firewalls have paid support available.

Readers, please think this over: Any properly configured firewall, even an open source firewall distribution, is certainly better than the best non-configured branded firewall!

SME network and security requirements

Let us consider the typical set-up of a small and medium business – a head office with 25 users, branch offices with 4–5 users, and a number of field representatives. Let us try to summarise their IT network requirements.

Head office:

- Computer systems network
- Isolation of Internet exposed systems from the internal network
- Internet access control – website filtering, services filtering
- Gateway anti-virus and anti-spam

Branch offices:

- Site to Site VPN Connectivity – for head office connectivity
- Internet access control – website filtering, services filtering

Field representatives:

- Host-to-network VPN connectivity (for travelling individuals)

Depending upon the number of users and the features required, the price for commercially available firewalls suitable for SME network ranges from Rs 10,000 to Rs 50,000. In addition, an annual subscription is required for continued support. The first few VPN client licences are free, while additional licences typically cost Rs 2000 per user. The firewall has a pre-defined lifecycle and one may have to purchase a new model after support for the current model expires. Against this, open source distros are available free and come with community support. Paid support/ maintenance is also available through local vendors.

IPCop 2.0.6, the open source firewall that runs on any old computer system and which meets most of these requirements, is simple to install and has excellent community

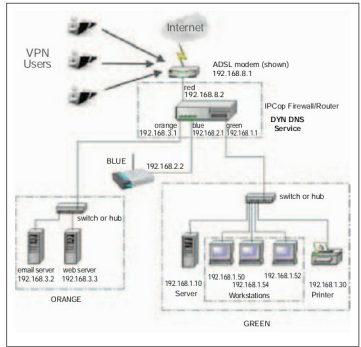


Figure 1: IPCop network

support, which is absolutely free!



Note: Gateway anti-virus and anti-spam module is currently not available for IPCop 2.0.6

The IPCop firewall

Let us try to understand a network that uses IPCop, with the help of a simple diagram (Figure 1).

At the head office, the IPCop firewall will segregate the internal network as:

Protected computers – Green network – 192.168.1.0

Protected wireless network – Blue network – 192.168.2.0

DMZ computers (systems to be accessed from the internal network and from the Internet) – Orange network – 192.168.3.0

Depending upon the number of users in any remote branch, the connectivity will be established either by site-to-site VPN (for over five users) or by site-to-host OpenVPN. These VPN users can access various services from the internal network—for example, a CRM/ERP software.

If more than five users from one branch wish to connect to the head office, at the same time, site-to-site VPN could be deployed using the IPCop firewall at that branch.

IPCop installation

Start by downloading the ISO image of the latest stable release from <http://ipcop.org/download.php>. Burn it to a CD. IPCop installation is very simple and straightforward. You require a computer system with a minimum of two Ethernet cards, 512 MB RAM, a 20 GB hard disk or a 1 GB flash drive, and a CD-ROM drive for installation. Start the installation by booting from the IPCop CD.

The first screen greets you with the IPCop mission